Thermoelectrically Cooled

HgCdTe Photodetector

Three-Dimensional Engineering Assessment

Revision 2 29 October 2026

Fine-mesh qualification: not qualified. Three converged fine meshes fail the unchanged independent external-bath raw-gradient heat-flux criterion of 1%, with errors of 1.51–1.66%. Coarse numerical passes and conservative reaction balance do not override this failure.

Abstract. A steady coupled finite-element model evaluates a three-stage thermoelectric cooler and HgCdTe detector within a nine-pin TO-66 enclosure. The construction contains 215 solid domains and all 88 thermoelectric legs. Actual tetrahedral boundaries and solved temperature and electric-potential fields support every geometry and field illustration. Gas heat transfer is a pressure-dependent surface-film reduction; diffuse-grey radiation and axial wire/lead operators are coupled to the solid solution. Ten coarse operating cases satisfy their per-case numerical checks. The illustrated nitrogen condition uses 109,771 nodes, 429,100 tetrahedra and 907 exchange patches, with a 300 K external header bath, 0.1 A drive and 2 mW detector deposition at 100 Pa. Its die volume mean is 268.806887 K, a result for declared hypotheses rather than a manufacturer-accurate or measured detector temperature.

Pre-solder at the header/alumina interface is physically present according to the supplied hardware information. The model uses a generic 50 μm, $k = 2$ W m^{-1} K^{-1} mount proxy; it is not a metallurgical pre-solder model. Material-grade identification, measured interfaces, spectral window transport, carrier transport and device validation remain incomplete.

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1 Model geometry

The solved construction combines a three-stage, 44-couple TEC with a Kovar-like header, finite-volume mount and die attach, HgCdTe die, Au pads, nine glass/pin pairs and an attached cap/window. A unique conductor path has 181 vertices, 180 edges and two external terminals. The 28/12/4 stage allocation, buried routes and ceramic supports are inferred construction, not verified manufacturer artwork. The package family envelope is supported by the public drawing [4].

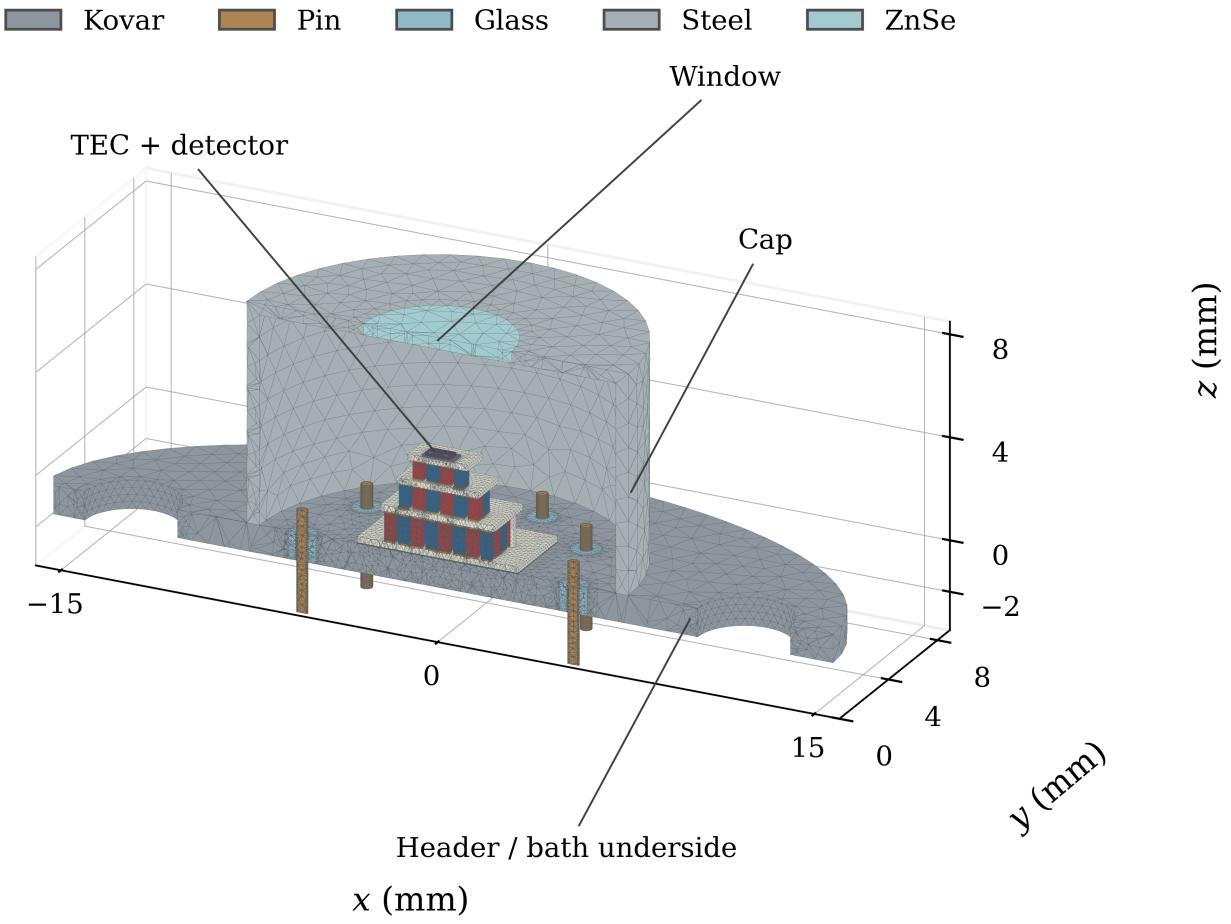


Figure 1: Actual package mesh cut at $y = 0$, retaining $y \geq 0$. Cut faces are closed by tetrahedron–plane intersections; material colours identify solids. The header, cap/window and central TEC remain their actual relative sizes. The view removes solids only for display and does not change the solved boundary conditions.

The header flange spans 31.5 by 17.5 mm. The cap outside diameter is 15.2 mm, with 8.6 mm height above the header. The window clear diameter is 5.9 mm. All displayed surfaces belong to the solved solid mesh. No gas-volume mesh or unresolved wire arc is drawn.

The **external header underside alone** is held at 300 K. Internal ceramics, pins, cap and window temperatures are unknowns. External cap/window surfaces and isolated pin ends are adiabatic. Cap/header and window/rim contacts impose ideal thermal continuity, without identified seal resistance.

1.1 Thermoelectric stack and detector

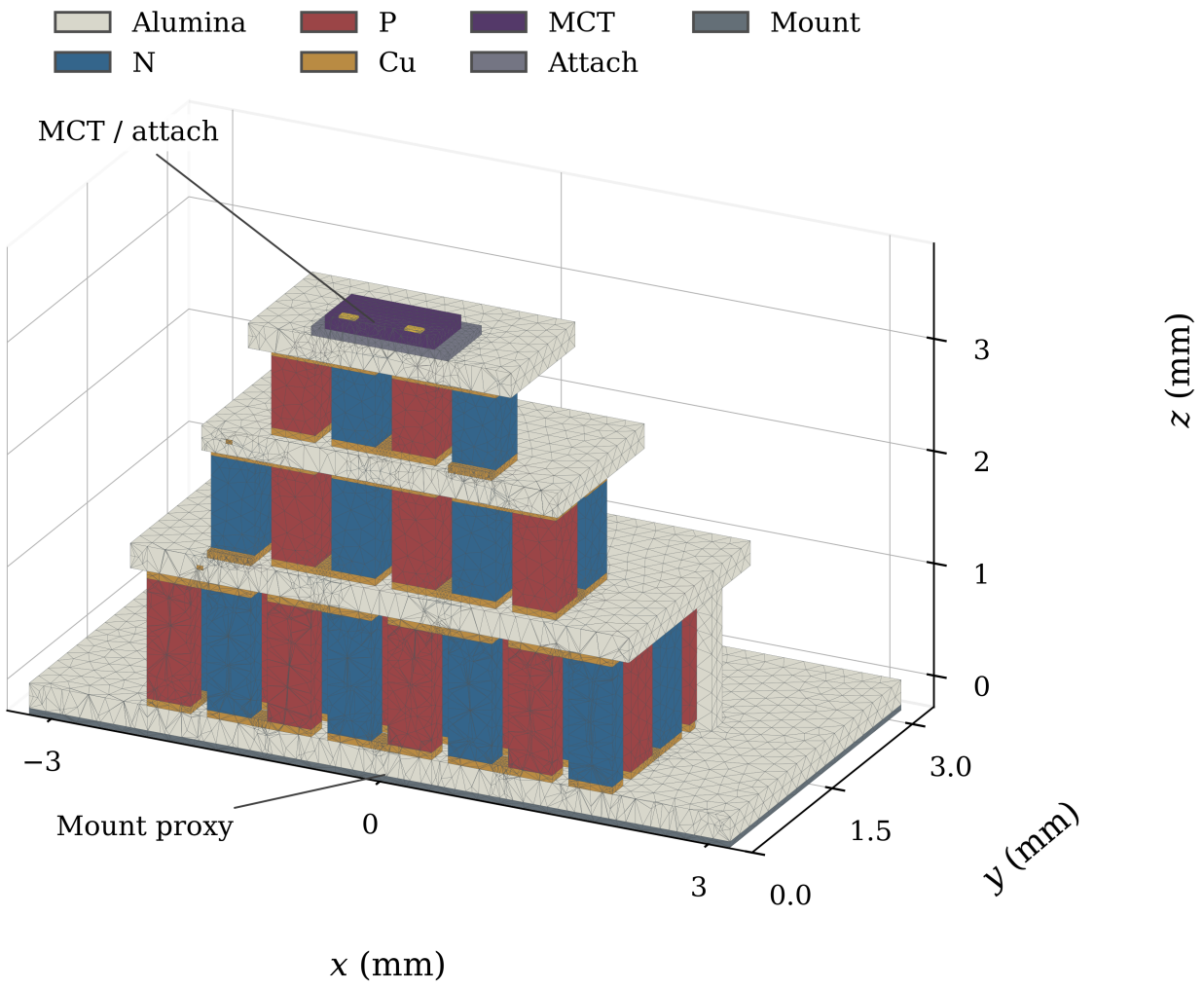


Figure 2: Enlarged actual TEC, mount, attach and detector mesh in a $y \geq 0$ cutaway. All three stages are visible. The mount underside is its real interface with the omitted header. P/N, Cu and alumina occupy their original material domains; no exposed external interstage jumpers are introduced.

Table 1: Resolved stack dimensions and reduced connections.

Part	Representation
TEC	Hot/cold widths 6.4/2.4 mm; nominal stack height 3.8 mm; 88 legs
Detector	1 mm square, 120 μm thick; 80 μm attach beneath
Header/alumina joint	50 μm, $k = 2$ W m^{-1} K^{-1} generic mount proxy
Fine Au bonds	Two axial operators, diameter 25 μm, length 7 mm
TEC supply leads	Two axial Cu operators, diameter 0.202 mm, length 8 mm

The Au/Cu operators use temperatures from their actual 3D endpoint regions. They conserve endpoint power but do not resolve wire shapes, spatial lead current, or distributed wire exchange with gas and radiation. Additional carriers, thermistor and exact manufactured detector-support layers are absent.

1.2 Hardware alignment and unresolved construction

Table 2: Physical/reference information versus the modeled construction.

Feature	Physical or drawing information	Modeled representation
Feedthrough layout	Approved reference depicts a C arrangement; exact pinout unknown	Nine generic, unnumbered electrical instances on a 10.8 mm pitch circle
Pin diameter	Family drawing: 0.76 mm [4]	0.44 mm; glass radius 0.55 mm
Supply leads	Illustrative tinned Cu diameter 0.28 mm	Hybrid Cu diameter 0.202 mm; constant transport proxies
Header/alumina pre-solder	**Physically present**; alloy, thickness and transport unqualified	Generic 50 μm mount with $k = 2$ W m^{-1} K^{-1}; not identified metallic pre-solder
Internal solder	Manufacturer Sn–Sb designation; artwork unverified	Ideal full-stack interfaces; earlier contact studies do not identify stack solder
Detector support	Actual supporting carrier stack unestablished	MCT and attach only; extra carriers and thermistor omitted
Window	5.9 mm nominal clear diameter; optical coating data absent	ZnSe candidate, internally opaque grey and externally adiabatic
Geometric contacts	Visual assembly clearances do not define thermal contacts	Conformal reconstructed solids with explicit interfaces

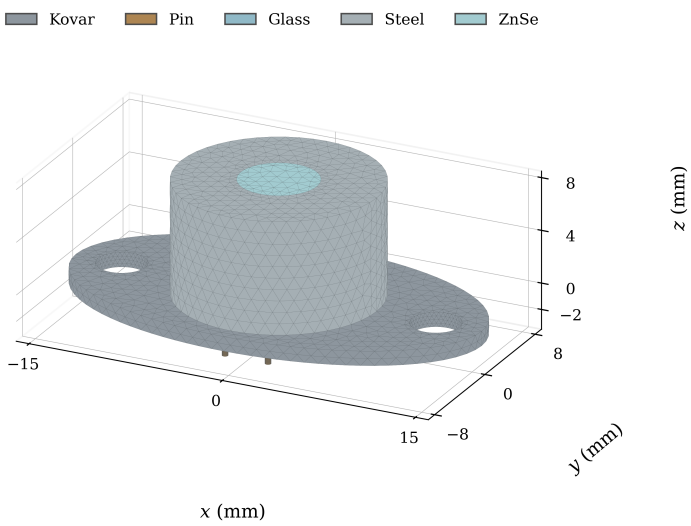


Figure 3: Actual exterior mesh. The opaque cap conceals the central cooler; this is not a transparent conceptual rendering.

Table 3: Actual pin centres (mm). Labels identify positions only, not electrical pin numbers.

Label	x	y
1	-5.400000	+0.000000
2	+5.400000	+0.000000
3	+4.368692	+3.174040
4	+1.668692	+5.135705
5	-1.668692	+5.135705
6	-4.368692	+3.174040
7	-4.368692	-3.174040
8	-1.668692	-5.135705
9	+1.668692	-5.135705

The listed coordinates preserve the solved geometry. A pitch-circle construction does not establish a measured C-layout, tooling tolerances or manufacturer pinout.

2 Governing physics

2.1 Coupled solids and boundary conditions

Elmer's coupled thermoelectric solver determines spatial temperature T and potential ϕ throughout the solid mesh. The sign convention is

$$\mathbf{E} = -\nabla\phi, \qquad \mathbf{j} = \sigma_e(\mathbf{E} - S\nabla T), \qquad \nabla\cdot\mathbf{j} = 0, \tag{1}$$

$$\mathbf{q} = -k\nabla T + ST\mathbf{j}, \qquad \nabla\cdot\mathbf{q} = \mathbf{j}\cdot\mathbf{E} + q'''_{\mathrm{det}}. \tag{2}$$

$\mathbf{j}\cdot\mathbf{E}$ is electrical work density; irreversible Joule heating is $|\mathbf{j}|^2/\sigma_e$. Ideal conformal contacts retain temperature and potential continuity. The TEC includes Seebeck, Peltier and Joule coupling. Its constant mixed verification law is not an identified manufacturer $S(T), \sigma_e(T), k(T)$ dataset.

Table 4: Operating condition of the illustrated enclosed-package solution.

Quantity	Specification
Thermal bath	300 K on external header underside only
TEC drive	0.1 A series current through the unique 88-leg path
Detector bias heat	Prescribed 1.5 mW over the MCT volume
Absorbed optical heat	Prescribed 0.5 mW over the same volume
Environment	Pure N_2, 100 Pa; diffuse-grey internal radiation
Contact model	Ideal solid interfaces plus finite mount/attach proxies
Time dependence	Steady solution; no calibrated transient response

The 2 mW detector deposition is one bias/optical thermal representation. A second detector Joule term is not added. The model does not solve carrier transport, bias-dependent absorption or thermistor electrical behavior.

Package solids use $S = 0$ and a tiny computational conductivity, $\sigma_e = 10^{-12}$ S m^{-1}. Their potential is a passive numerical extension, not real Kovar/Au electrical transport. Electrical figures therefore interpret P/N/Cu only. Spurious package Joule heat stays below 1 nW, and a separate 100-fold conductivity reduction changes the die mean by less than 5×10^{-7} K in the tested coarse case.

2.2 Hybrid wire and lead paths

Each axial connection has $G = kA/L$. Au bonds use $k = 315$ W m^{-1} K^{-1} and equal/opposite thermal powers. Cu leads use $k = 400$ W m^{-1} K^{-1} and $\rho_e = 1.72 \times 10^{-8}$ Ω m, with $R = \rho_e L/A$. For a lead connecting two solved endpoint temperatures, the two inward powers are

$$P_1 = G(T_2 - T_1) + \tfrac{1}{2}I^2R, \qquad P_2 = G(T_1 - T_2) + \tfrac{1}{2}I^2R. \tag{3}$$

Thus axial conduction cancels in the whole-package budget; the lead electrical work remains. These conservative operators update during the nonlinear solid solve. They are a 3D-solid/1D-wire hybrid, not an all-3D wire model.

2.3 Surface radiation and pressure-dependent gas paths

Diffuse-grey radiosity is coupled nonlocally to the current surface temperatures:

$$J_i^{\mathrm{rad}} = \epsilon_i \sigma_{\mathrm{SB}} T_i^4 + (1-\epsilon_i) \sum_j F_{ij} J_j^{\mathrm{rad}}, \tag{4}$$

$$P_i^{\mathrm{rad,in}} = -A_i \left(J_i^{\mathrm{rad}} - \sum_j F_{ij} J_j^{\mathrm{rad}} \right), \tag{5}$$

$$\sum_j F_{ij} = 1, \qquad A_i F_{ij} = A_j F_{ji}. \tag{6}$$

Deterministic first-hit rays account for occlusion. Symmetric area balancing preserves blocked-pair zeros. Balanced closure below 10^{-10} and reciprocal area error below 10^{-15} m^2 establish conservation, not exact view factors. For 907 patches, raw reciprocity L1/area is 0.213 and raw area-weighted closure error is 0.0136. Finite-ray sampling, patch resolution and hit mapping limit accuracy.

The reduced gas-film network uses symmetric exchange area $W_{ij} = A_i F_{ij}$ and harmonic normal-projected first-hit flight gap L_{ij}:

$$G_{ij} = \frac{k_g(\overline{T}_{ij}) W_{ij}}{L_{ij} + 2\ell_T}, \qquad P_i^{\mathrm{gas,in}} = \sum_j G_{ij}(T_j - T_i), \tag{7}$$

$$\lambda = \frac{\mu}{p}\sqrt{\frac{\pi R_g T}{2}}, \qquad \ell_T = \frac{2-\alpha}{\alpha}\frac{2\gamma}{\gamma+1}\frac{\lambda}{\mathrm{Pr}}, \tag{8}$$

$$\mathrm{Kn} = \lambda / L, \qquad \mu = \frac{k_g}{C_p + 1.25 R_g}. \tag{9}$$

Here R_g is the specific gas constant and $\alpha = 0.8$ is an accommodation hypothesis. Eucken viscosity and constant ideal-gas C_p, γ are closures. The implementation gives exactly zero gas exchange at $p = 0$ and recovers parallel-plane continuum and Kennard free-molecular limits [6]. This is a reduced thermal path, not a gas-volume conduction PDE or CFD solution.

Table 5: Ingested dilute-gas conductivity, W m^{-1} K^{-1} [1].

Gas	200 K	300 K	400 K
N_2	.0183	.0259	.0326
Ar	.0124	.0177	.0224
Kr	.0064	.0095	.0122

Interpolation is rejected outside 200–400 K. At 100 Pa, bounding Rayleigh numbers are below one for 1, 3, 8 and 14.2 mm gaps at 250 K with a 100 K temperature difference. Atmospheric krypton flags further convection assessment; 1708 is a heated-below horizontal reference, not a universal cavity threshold.

The thermal window is internally opaque grey ($\tau = 0$) and externally adiabatic. Spectral LWIR transmission, optical coatings and ambient radiance are absent. Prescribed absorbed optical heat does not predict transmission through the window.

3 Operating response

3.1 Three-dimensional thermal field

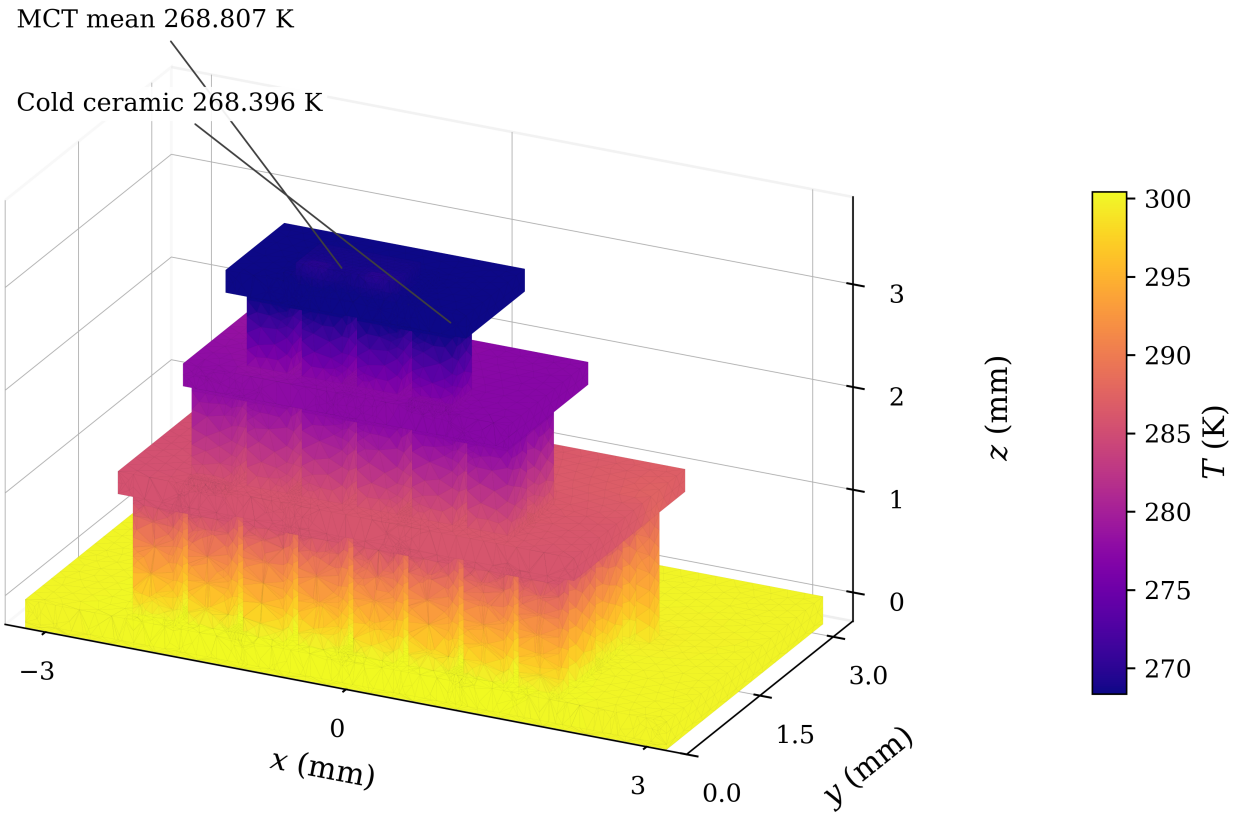


Figure 4: Actual solved temperature on the true $y \geq 0$ TEC/attach/MCT cutaway. Boundary triangles use their mean P1 temperature; cut-face vertices carry exact mesh-edge interpolants. Annotations report actual solid volume means, not temperatures sampled at arrow tips. $I = 0.1$ A, $T_b = 300$ K, $p = 100$ Pa N_2; detector heat 2 mW.

Table 6: Representative coarse solution with 907 exchange patches.

Quantity	Value
MCT die volume mean	268.806887 K
Cold ceramic volume mean	268.395968 K
TEC terminal voltage	0.537982496 V
Independent bath-gradient error	0.420150%

The 0.411 K die-to-cold-ceramic mean difference includes finite attach conduction, spreading and parasitic paths. A smooth temperature field does not establish independent heat-flux accuracy. The fine-mesh qualification failure remains applicable to the enclosed-package model.

3.2 Detector detail and exact section

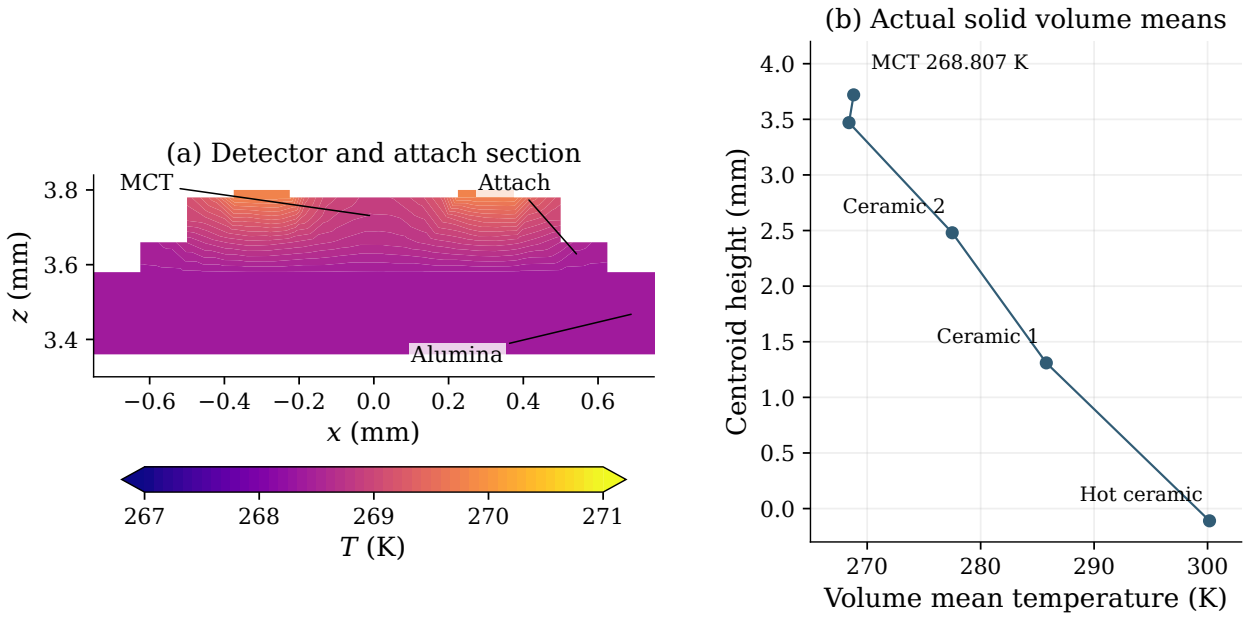


Figure 5: Enlarged solved detector/attach cross-section and actual solid volume means. Panel (a) uses bounded P1 interpolation only within intersected tetrahedra; panel (b) joins sampled body means to guide the eye and is not a continuous axial temperature profile. $I = 0.1$ A, $T_b = 300$ K, $p = 100$ Pa N_2; detector heat 2 mW.

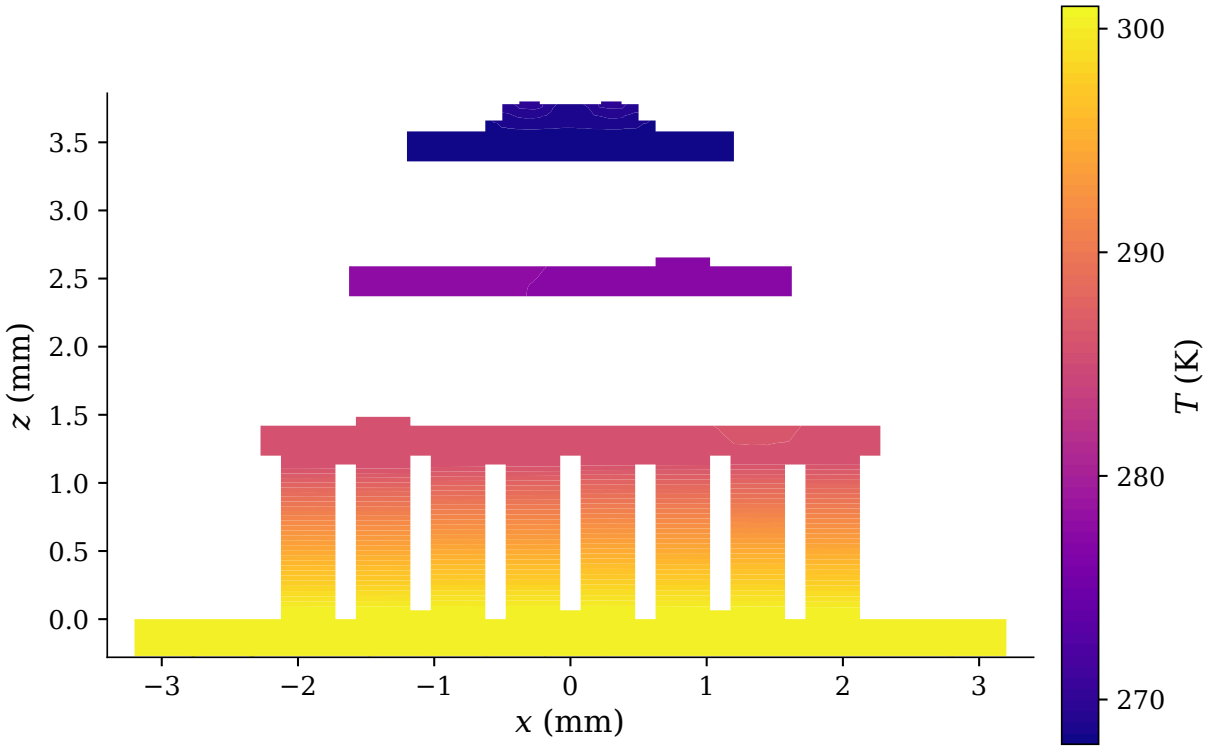


Figure 6: Actual $y = 0$ temperature section through the cooler and detector. The plane intersects 8 of 88 leg domains; upper-stage rows lie off the plane. White gaps are actual unmeshed voids, not missing temperature interpolation. $I = 0.1$ A, $T_b = 300$ K, $p = 100$ Pa N_2; detector heat 2 mW.

The plane yields 12,160 exact intersection triangles; coincident faces use a positive-side trace. Triangle connectivity preserves the solid voids.

3.3 Electrical and heat-flow diagnostics

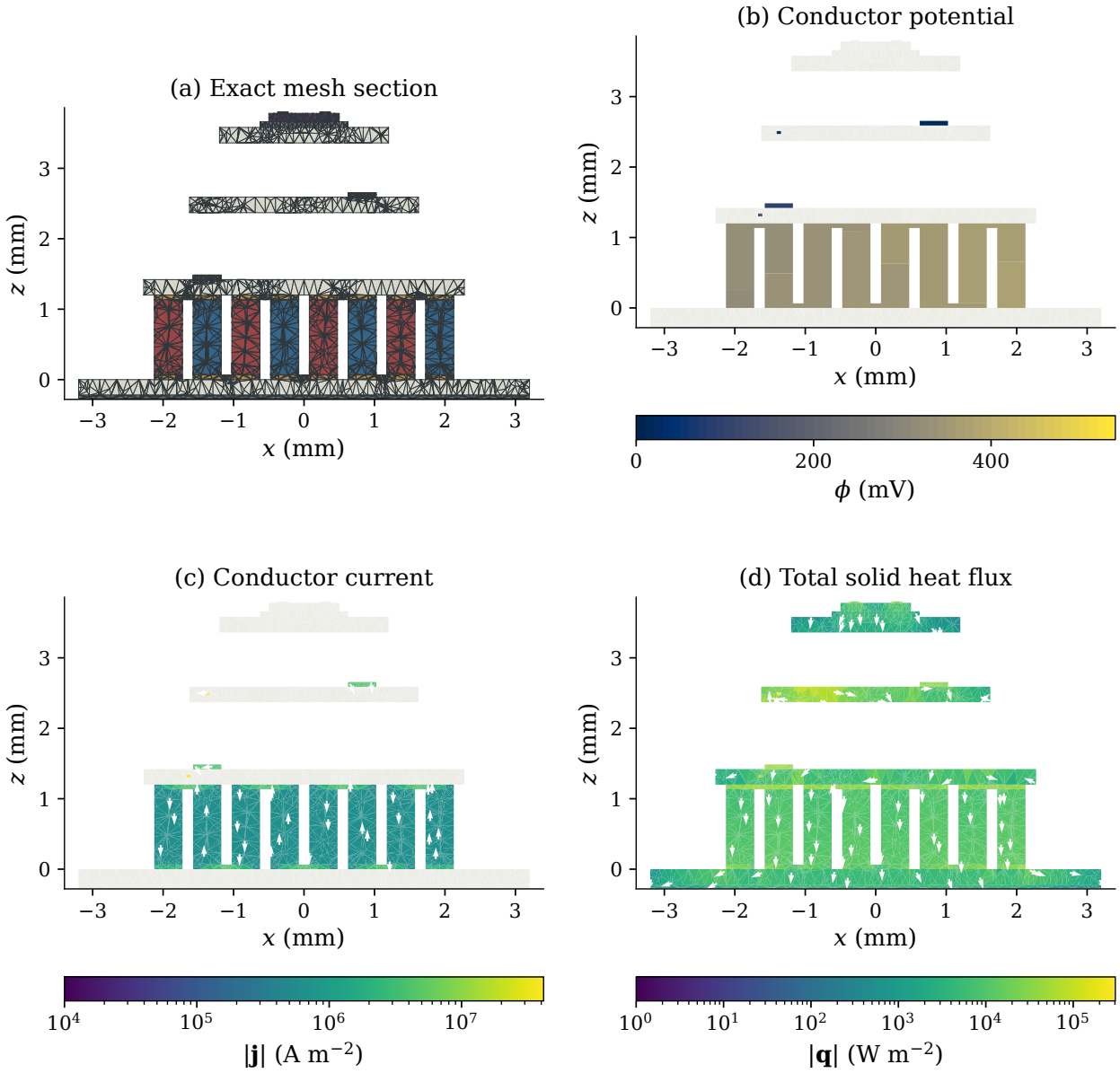


Figure 7: Registered actual-mesh section: (a) tetrahedral intersections, (b) potential on P/N/Cu only, (c) cell-mean conductor current density, and (d) cell-mean total solid heat flux including $ST\mathbf{j}$. Arrows encode projected direction; logarithmic colour encodes vector magnitude. Grey electrical regions are outside physical interpretation. Heat flux below 1 W m^{-2} is grey. $I = 0.1$ A, $T_b = 300$ K, $p = 100$ Pa N_2; detector heat 2 mW.

Raw and derived nodes are matched bijectively within 10^{-12} m; actual cells and material assignments agree. Reconstructed $\mathbf{j}$ and $\mathbf{q}$ from raw T, ϕ differ from the displayed derived vectors by normalized maxima `3.771e-16` and `3.098e-15`. These are data-correspondence checks, not physical error estimates. Local sharp-corner peaks remain unqualified.

3.4 Environment and lead comparisons

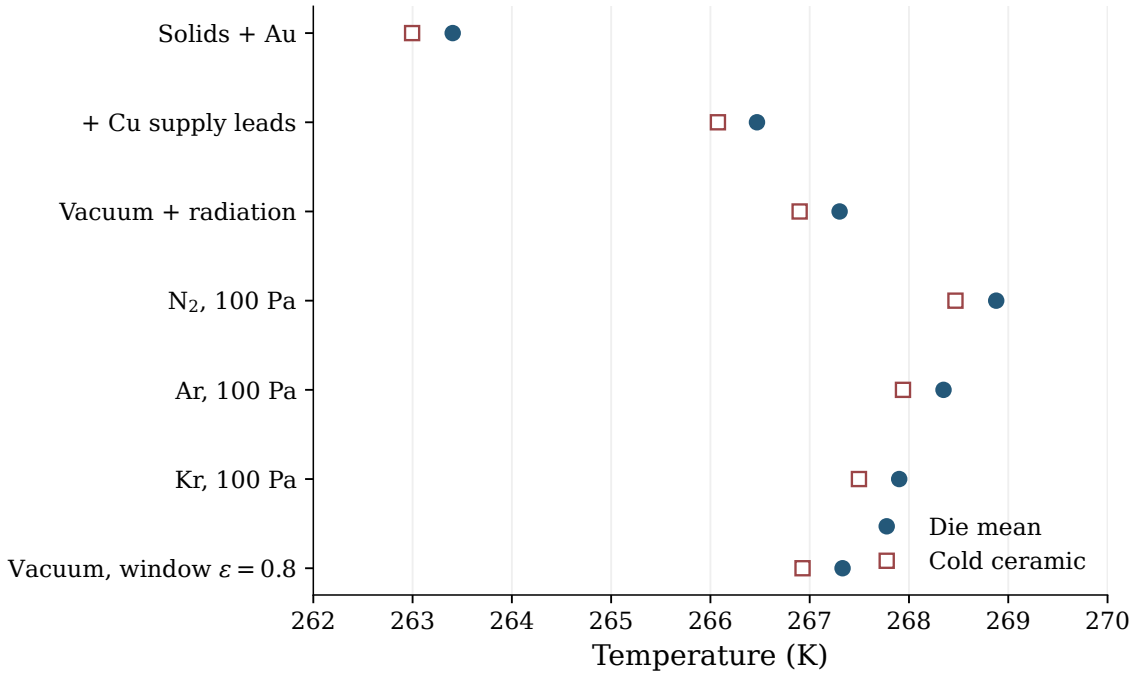


Figure 8: Actual accepted coarse operating points at 0.1 A, 300 K external bath and 2 mW detector heat. Exchange-enabled comparisons use 442 patches. Pure-gas rows are at 100 Pa; the final window-emissivity sensitivity is in vacuum. Fixed-current temperatures do not establish minimum power at a fixed detector target.

Table 7: Accepted coarse comparisons. Bath error is the signed independent gradient result.

Case	Cold (K)	Die (K)	TEC (V)	Bath error (%)
Solids + Au	262.995146	263.404041	0.566165120	-0.35367
+ Cu supply leads	266.075552	266.468861	0.547730485	+0.84352
Vacuum + radiation	266.898609	267.301094	0.544432801	+0.68673
N_2, 100 Pa	268.468131	268.878638	0.537501409	+0.41736
Ar, 100 Pa	267.939550	268.347858	0.539885636	+0.50418
Kr, 100 Pa	267.496013	267.901465	0.541796372	+0.58561
Vacuum, window $\epsilon = 0.8$	266.927891	267.330990	0.544314237	+0.67075

The representative 907-patch nitrogen solution has die mean 268.806887 K, compared with 268.878638 K for 442 patches on the same bulk mesh. This 0.071751 K shift is a surface-resolution effect, not material calibration.

The earlier bare-package fine case gave die 263.389718 K and cold ceramic 262.965994 K. It has 199 domains and two pins, without cap/window, gas, radiation or Cu leads. It is a separate geometry and is not merged with the enclosed-package fields. None of these comparisons validates manufacturer accuracy.

3.5 Whole-package energy and internal transfers

$$Q_{\text{header}} = IV_{\text{TEC}} + P_{\text{bias}} + P_{\text{optical}} + \sum_{\ell \in \text{Cu leads}} I^2 R_\ell. \tag{10}$$

The illustrated solution has $Q_{\text{header}} = 55.884122784$ mW, $IV_{\text{TEC}} = 53.798249618$ mW and lead Joule heat 0.085872837 mW. Detector deposition is 2 mW. The conservative global residual is `3.292e-10` W; this reaction balance is distinct from direct-gradient accuracy. Radiation, gas and

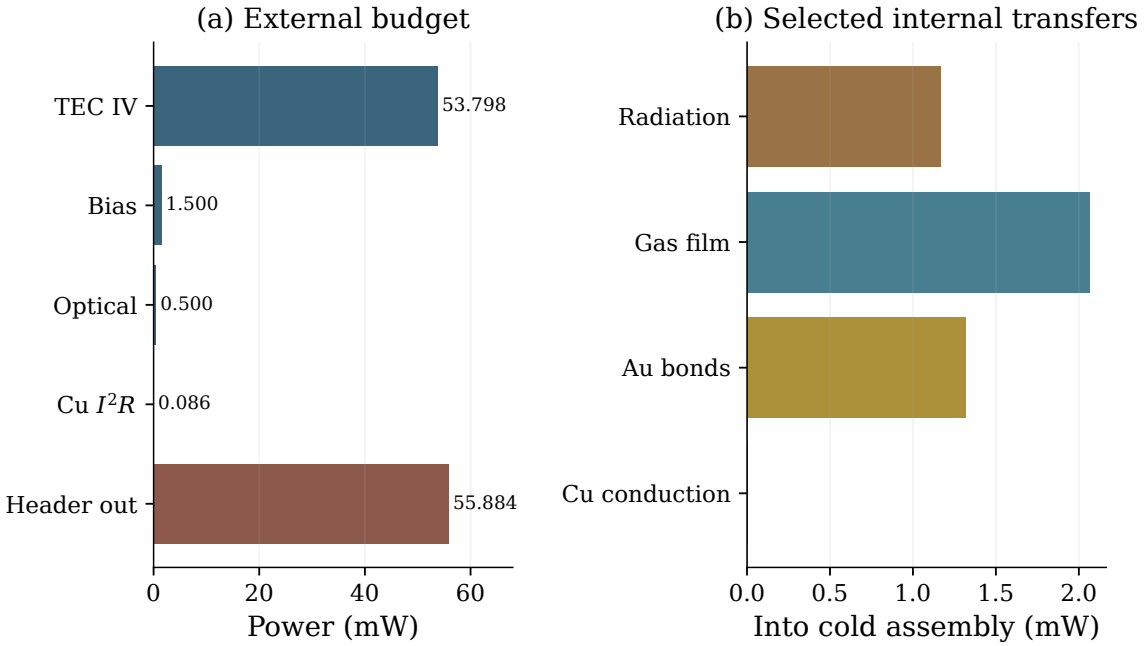


Figure 9: (a) External electrical/deposited powers and conservative bath reaction. (b) Selected signed internal transfers into the die/attach/pads/cold-ceramic group. Other TEC surfaces are excluded from panel (b). Internal transfers are not added again as external input. $I = 0.1$ A, $T_b = 300$ K, $p = 100$ Pa N_2; detector heat 2 mW.

Table 8: Signed inward internal powers by solid group (mW).

Group	Radiation	Gas	Au	Cu axial	Cu Joule
Die/attach/Au pads	+0.221092	+0.312160	+1.317957	+0.000000	+0.000000
Cold ceramic	+0.942649	+1.749731	+0.000000	+0.000000	+0.000000
Cap/window	-2.463572	-3.989961	+0.000000	+0.000000	+0.000000
Nine pins	-0.018260	+0.112985	-1.317957	-17.714395	+0.042936
Other solids	+1.318091	+1.815085	+0.000000	+17.714395	+0.042936

axial Au/Cu conduction cancel over the complete assembly. Each Cu lead contributes its electrical work once. Whole-package horizontal control volumes include cap, pins and nonlocal sources, so they are not isolated textbook TEC-stage balances. Cap/flange power inferred from a surface budget is conservative but is not an independent cap-gradient measurement.

4 Mesh & verification

4.1 Verification matrix

PASS applies only to the stated test and scope. Ten coarse integrated executions pass their per-case gates. Three genuine converged fine executions fail the independent bath-gradient test. The full enclosed-package numerical qualification is therefore **not qualified**, and device accuracy is unqualified.

Table 9: Verification matrix. Maxima refer to accepted coarse cases unless a fine-mesh or independent reference scope is stated.

Test	Criterion	Evidence / scope	**Verdict**
Mesh / topology	Positive cells; conformal contacts; unique two-terminal path	215 domains; 88 legs; 181 vertices / 180 edges	**PASS**
Field correspondence	Bijective nodes $\leq 10^{-12}$ m; exact cells / bodies	Raw and derived fields agree; $\mathbf{j}, \mathbf{q}$ reconstructed	**PASS**
Interior current cuts	$< 0.5\%$ driven; $< 1\,\mu$A at zero drive	175 cuts; driven maximum 0.249%	**PASS**
Current balance	$< 10^{-6}$ A	Maximum 1.31×10^{-9} A	**PASS**
Free current residual	$< 10^{-6}$ A	Maximum 3.76×10^{-8} A	**PASS**
Free heat residual	$< 10^{-6}$ W	Maximum 6.30×10^{-9} W	**PASS**
Global energy balance	$< 10^{-6}$ W	Maximum 5.51×10^{-10} W	**PASS**
Coarse bath gradient	$< 1\%$ independent direct flux	Worst accepted case 0.844%	**PASS**
Independent plate solves	$< 10^{-6}$ K; heat balance $< 10^{-9}$ W	Two Elmer references; maximum 3.99×10^{-8} K	**PASS**
Internal exchange balance	$< 10^{-9}$ W per mechanism	8 exchange-operator cases; radiation, gas, Au/Cu	**PASS**
Surface refinement	$< 1\%$ of lift; $< 1\%$ voltage change	442 to 907 patches: 0.228% lift; 0.089% voltage	**PASS**
Fine bath gradient	$< 1\%$ independent direct flux	Three converged meshes: 1.51–1.66%	**FAIL**
Fine-mesh qualification	Every declared numerical gate must pass	Independent bath-gradient criterion fails	**NOT QUALIFIED**
Manufacturer / device accuracy	Matched materials, interfaces and independent device data	No qualified full-device comparison	**NOT QUALIFIED**

The two independent references solve 1D conducting plates with the same compiled radiation/gas routine, and compare with analytic parallel-plane heat balances. They check implementation limits, not the accuracy of a 3D enclosure view factor. A failed initial plate-startup test is retained as failed evidence; two corrected Elmer references support the reported passes.

4.2 Independent fine-mesh failure

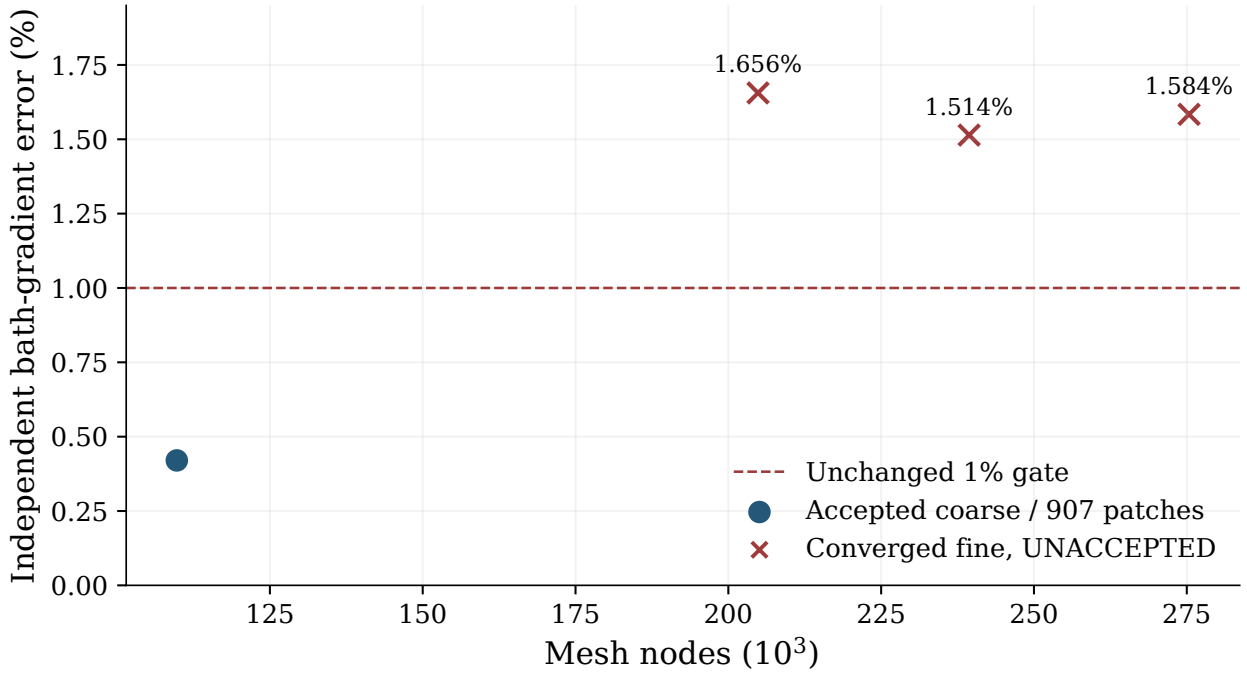


Figure 10: Independent external-header raw-gradient heat-flux error. The representative coarse solution passes; all three fine solutions exceed the unchanged 1% criterion. Solver convergence and small temperature/voltage changes do not waive this failure.

Table 10: Converged fine solutions that fail qualification.

Refinement	Nodes	Tetrahedra	Bath error (%)
Global refinement	204,855	821,874	1.655873
Header-bath refinement	239,394	967,731	1.514122
Feedthrough refinement	275,377	1,194,385	1.583835

All three fine executions converge and pass their other numerical gates. The independent bath-gradient errors are 1.655873%, 1.514122% and 1.583835%. Boundary/interface localization found the largest discrepancy near the cold-return feedthrough. Header-bath edge refinement and a separate glass/pin/header study were insufficient. The cause remains unresolved.

Conservative reactions are not substituted for the failed independent diagnostic. Divergent iterations, a timeout, invalid cross-mesh potential initialization and an over-budget mesh proposal remain rejected evidence. No numerical tolerance has been relaxed to improve the verdict.

Permitted interpretation. The coarse solutions support implementation checks and conditional engineering comparisons. They do not establish complete mesh convergence, a resolved local corner flux, or device-level predictive accuracy.

5 Materials and uncertainty

Table 11: Steady thermal inputs and their qualification limits.

Input	Value / coverage	Classification
MCT	$Hg_{0.71}Cd_{0.29}Te$; $k = 2.41, 1.61, .973, .790$ at 90,148,298,400 K	Primary Table 2 example [2]; bounded local interpolation, actual detector grade unidentified
TEC	Constant mixed P/N coefficients, below	Verification law, not manufacturer-matched temperature-dependent BiTe
Alumina / Cu	$k = 30/400$ W m^{-1} K^{-1}	Frozen verification proxies
Kovar / Au	$k = 17.3/315$ W m^{-1} K^{-1}	Grade and low-temperature dependence unmeasured
Glass / mount / attach	$k = 1/2/1$ W m^{-1} K^{-1}	Construction hypotheses, not measured joints
Cap / window	$k = 15/18$ W m^{-1} K^{-1}	Stainless-like cap; ZnSe vendor proxy at 298 K [3]
Gas	Dilute $k(T)$, 200–400 K	Source-backed [1]; film geometry and accommodation approximate
Emissivity	P/N .3; alumina/MCT/glass/mount/attach .8; Cu/Au .05; header/pins .1; cap .2; window .1/.8	Diffuse-grey hypotheses, not measured coatings

Table 12: Constant P/N verification coefficients.

Material	k (W m^{-1} K^{-1})	σ_e (S m^{-1})	S (μV K^{-1})
P	1.5	1.400×10^5	+160
N	1.5	1.655×10^5	−170

Pre-solder is physically present at the header/alumina interface. The 50 μm, $k = 2$ W m^{-1} K^{-1} proxy gives $R'' = t/k = 25 \times 10^{-6}$ m^2 K W^{-1} before spreading. It does not identify an alloy, wetting layer, intermetallic structure or measured joint resistance. Replacing it requires new physical inputs and new verification.

The MCT source reports conductivity uncertainty of approximately 7–14%; this is not a package-temperature confidence interval. No composition interpolation, material stitching or out-of-domain extrapolation is used. The gas source gives generally below 3% conductivity uncertainty, which also does not include the surface-film reduction or package construction uncertainty.

Geometry, material grade, interfaces, enclosure heat paths and manufacturer calibration remain confounded. Constant coefficients, opaque window assumptions and steady heat-capacity placeholders do not qualify low-temperature material curves, optical responsivity or transients. Display precision is numerical precision, not an empirical temperature uncertainty.

6 Recommendations

1. Resolve the independent header-bath gradient failure using systematic boundary/interface refinement or an independently verified discretization. Retain the 1% gate and report reaction and gradient diagnostics separately.
2. Identify the actual header/alumina pre-solder, die attach and seal construction, including alloy, thickness, contact area and thermal resistance. Replace generic proxies only with new, separately verified cases.
3. Obtain matched-grade, temperature-dependent P/N transport and independent loaded device measurements. Reserve calibration and holdout conditions before fitting; require matched bath, drive, detector load and environment.
4. Characterize the real feedthrough arrangement, wire lengths and optical window. Treat spectral transmission, external radiance and distributed wire gas/radiation exchange as additional physics requiring verification.
5. Compare electrical power at a common detector target after numerical and device qualification. Fixed-current gas comparisons alone do not establish a production gas choice, optimum cooling performance or reliability.

All figures here derive from existing solved fields. This assessment revises presentation only; it introduces no new physical execution or acceptance claim.

7 References

1. NIST/CRC, *Thermal Conductivity of Gases*, 2019. Selected dilute-gas table, 200–400 K.

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2. *Applied Optics* 47, 578 (2008), primary Table 2, $Hg_{0.71}Cd_{0.29}Te$ thermal-conductivity example. DOI: 10.1364/AO.47.000578.

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3. Crystran, *Zinc Selenide (ZnSe)*; vendor thermal-conductivity proxy at 298 K.

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